THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

Dec 16, 2021 09:30 pm – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) With the aid of suitable diagram(s), explain the mechanism of "design and build" and "management contracting". [15 marks]

   b) While procurement method is a means to distribute the risks, highlight the risks undertaken by the parties involved in each of the two procurement methods in (a) above. [10 marks]

2. a) An investor is planning to build a 200-unit service apartment. The usable floor area of this service apartment will be 9,000m^2 while the circulation area will take up 20% of the total floor space. The construction cost is estimated to be \$10,000/m^2 and the building will take 2 years to construct. At a monthly rent of \$25,000/unit, the service apartment will attract 90% occupancy during its first three years of operation before attaining a full occupancy rate from the fourth year onward. The initial yield for this type of development would be 6%. With the cost of advertisement being \$1,000,000 and a short-term finance of 12% per annum. By considering the developer's profit (15%), professional fee (13%), contingency (5%) and acquisition costs (4%), calculate the maximum site value based on the above parameters and your engineering judgement. [18 marks]

   b) Briefly describe the major drawbacks of residual method and under what circumstances should discount cash flow methods be used for investment appraisal. [7 marks]

3. A client is interested to invest in either Scheme 'A' or Scheme 'B', and the projected cash flow (all values in \$ million) of the two schemes is as follows:

   Scheme 'A'
   Investment period: 8 years
   Capital cost: \$100
   Spending: \$15 in year 5 (for maintenance); \$5 in year 7 (for maintenance)
   Income in years 1-8: \$30, \$25, \$20, \$15, \$20, \$18, \$16 and \$15
   Reversion value: \$35

   Scheme 'B'
   Investment period: 10 years
   Capital cost: \$100
   Spending: \$8 in year 6 (for maintenance)
   Income in years 1-10: \$20, \$22, \$24, \$22, \$20, \$22, \$20, \$18, \$16 and \$16
   Reversion value: \$0

   a) Using a discount rate of 8%, calculate the net present value of both Schemes 'A' and 'B'. [10 marks]

   b) Compute the internal rate of return of both Schemes 'A' and 'B', and determine which is a more attractive scheme based on the internal rate of return [15 marks]

4. The following table shows the duration and logical relationship of the activities involved in a construction project:

Activity ID	*Predecessor*	*Duration (days)*
A	--	10
B	A	30
C	B	30
D	C (SS10)	40
E	D (SS10)	40
F	C	40
G	D	28
H	E	30
I	D	30
J	E	30
K	G, H, I, J	70
L	D, E, F	60
M	K	100
N	L, M	10

a) Prepare an activity-on-node network diagram based on the information given in the table above and identify the critical path of this project. [12 marks]

c) Prepare a table showing the early and late start and completion dates of each activity and work out the total and free floats accordingly. [8 marks]

d) Describe what measures should be taken by the project manager to monitor and control project progress. [5 marks]

5. a) A Hong Kong based construction company is keen to participate in projects along the Belt and Road regions. Discuss whether it is suitable to adopt a functional organisation approach when branch offices are first established in the Belt and Road countries for tendering purpose. [10 marks]

b) Explain the characteristics of a project based and matrix organisation and discuss under what circumstance the construction company should consider employing these two organisational settings. [15 marks]

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